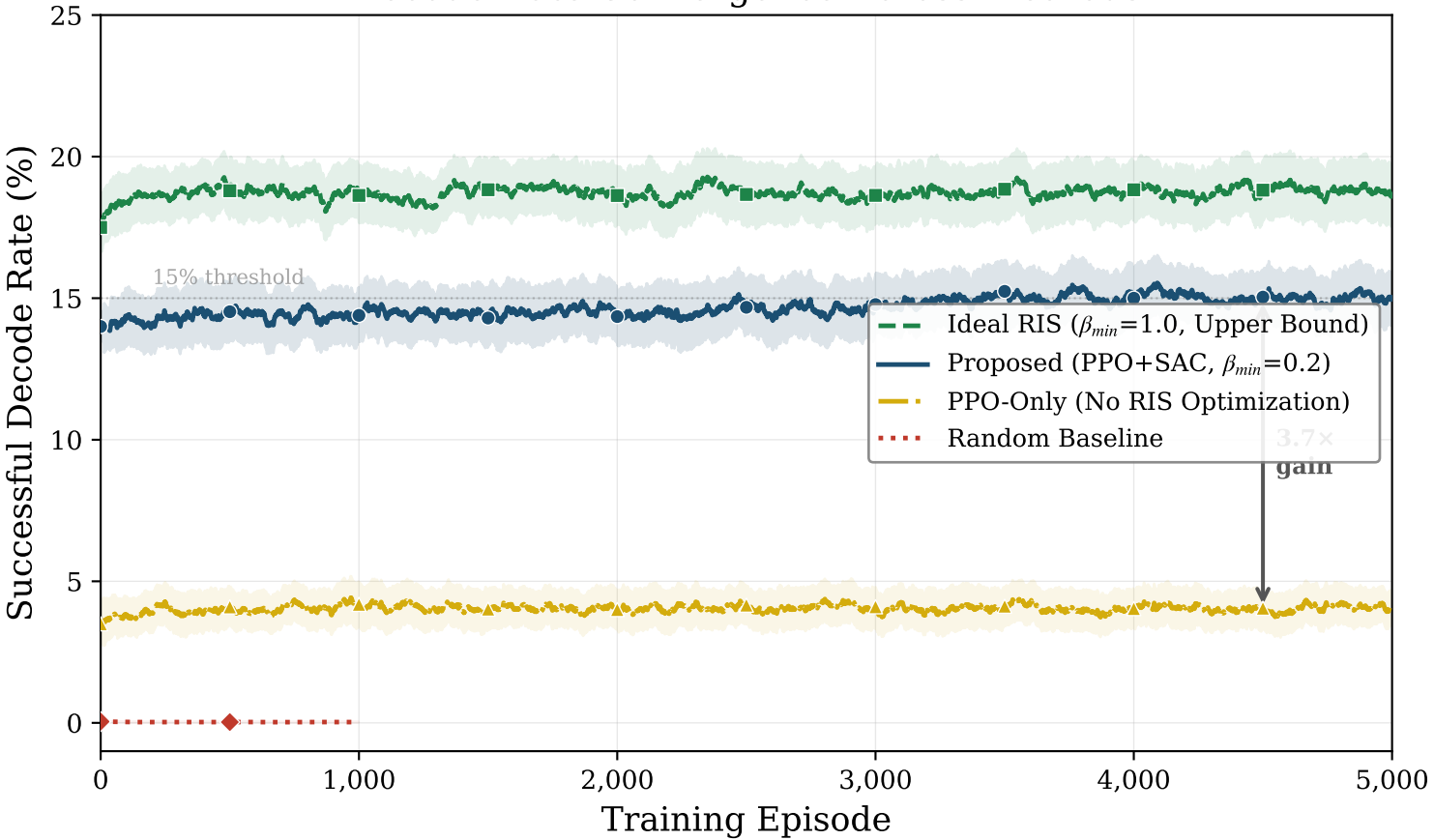

Decode Rate Convergence Across Methods
Successful Decode Rate (%)
Training Episode
25
20
15
10
5
0
0
1,000
2,000
3,000
4,000
5,000
15% threshold
Ideal RIS (β_{min}=1.0, Upper Bound)
Proposed (PPO+SAC, β_{min}=0.2)
PPO-Only (No RIS Optimization)
Random Baseline
3.7×
gain